MIT

The function $T = x^2 + 2y^2 + 2z^2$ gives the temperature at each point in space.

1) At the point $P = (1, 1, 1)$, in which direction should you go to get the most rapid decrease in T? What is the directional derivative in this direction?

The most rapid decrease at P occurs in the direction of $-\nabla T|_P$.

$$\begin{aligned} -\nabla T|_P &= -\left\langle 2x, 4y, 4z \right\rangle\Big|_{(1,1,1)} \\ &= -\langle 2, 4, 4 \rangle \end{aligned}$$

The unit vector in this direction is simply $-\frac{1}{\sqrt{36}}\langle 2, 4, 4\rangle = \boxed{\left\langle -\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3} \right\rangle} = \hat{\mathbf{u}}$.

The directional derivative at P in this direction is given by $\mathrm{D}_{\mathbf{u}} T|_P$:

$$\begin{aligned} \mathrm{D}_{\mathbf{u}} T|_P &= \nabla T|_{(1,1,1)} \cdot \left\langle -\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3} \right\rangle \\ &= \langle 2, 4, 4 \rangle \cdot \left\langle -\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3} \right\rangle \\ &= -\frac{2}{3} - \frac{8}{3} - \frac{8}{3} \\ &= \boxed{-6} \end{aligned}$$

2) At P, about how far should you go in the direction found in part (1) to get a decrease of 0.3?

T decreases at an instantaneous rate of 6 units of temperature per unit of distance. Hence, traveling $\frac{0.3}{6} = \boxed{0.05}$ units along $\mathbf{u}$ will result in an approximate decrease of 0.3 in T.

Sinclair, Problem 1

Find the directional derivative of f in the same direction as v at the point indicated.

a) $f(x, y) = 4x + 3y$, $\mathbf{v} = \langle 1, -1 \rangle$, (x, y)

$$\begin{aligned} \mathrm{D}_{\mathbf{v}} f &= \nabla f \cdot \hat{\mathbf{v}} \\ &= \langle 4, 3 \rangle \cdot \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle \\ &= \frac{4}{\sqrt{2}} - \frac{3}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}} \end{aligned}$$

b) $f(x, y) = x^2 + y^2$, $\mathbf{v} = \langle 3, 1 \rangle$, $(1, -1)$

$$\begin{aligned} \mathrm{D}_{\mathbf{v}} f &= \nabla f\Big|_{(1,-1)} \cdot \hat{\mathbf{v}} \\ &= \langle 2, -2 \rangle \cdot \left\langle \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right\rangle \\ &= \frac{6}{\sqrt{10}} - \frac{2}{\sqrt{10}} = \frac{4}{\sqrt{10}} = \boxed{\frac{2\sqrt{10}}{5}} \end{aligned}$$

c) $f(x,y,z) = x^2 + y^2 - z$**,** $\mathbf{v} = \langle 1,1,1\rangle$**,** $(-1,1,-1)$

$$\begin{aligned} D_{\mathbf{v}} f &= \nabla f\big|_{(-1,1,-1)} \cdot \hat{\mathbf{v}} \\ &= \langle 2x, 2y, -1\rangle\big|_{(-1,1,-1)} \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle \\ &= \langle -2, 2, -1\rangle \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle \\ &= -\frac{2}{\sqrt{3}} + \frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}} = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}} \end{aligned}$$

d) $f(x,y,z) = 3e^{xyz}$**,** $\mathbf{v} = \langle 1,1,1\rangle$**,** $(2,1,-1)$

$$\begin{aligned} D_{\mathbf{v}} f &= \nabla f\big|_{(2,1,-1)} \cdot \hat{\mathbf{v}} \\ &= \left\langle 3yze^{xyz}, 3xze^{xyz}, 3xye^{xyz} \right\rangle\big|_{(2,1,-1)} \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle \\ &= \left\langle -3e^{-2}, -6e^{-2}, 6e^{-2} \right\rangle \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle \\ &= -\frac{3e^{-2}}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{e^2}} \end{aligned}$$

Sinclair, Problem 2

Find the gradient of f **at the given point.**

a) $f(x,y) = 3x^2 + y^2$**,** $(2,0)$

$$\begin{aligned} \nabla f\big|_{(2,0)} &= \langle f_x, f_y\rangle\big|_{(2,0)} \\ &= \langle 6x, 2y\rangle\big|_{(2,0)} \\ &= \boxed{\langle 12, 0\rangle} \end{aligned}$$

b) $f(x,y) = 4x^2 - 5y^2$**,** $(1,1)$

$$\begin{aligned} \nabla f\big|_{(1,1)} &= \langle f_x, f_y\rangle\big|_{(1,1)} \\ &= \langle 8x, -10y\rangle\big|_{(1,1)} \\ &= \boxed{\langle 8, -10\rangle} \end{aligned}$$

c) $f(x,y) = \sin\frac{x}{y}$**,** $(\pi,4)$

$$\begin{aligned} \nabla f\big|_{(\pi,4)} &= \langle f_x, f_y\rangle\big|_{(\pi,4)} \\ &= \left\langle \frac{1}{y}\cos\frac{x}{y}, -\frac{x}{y^2}\cos\frac{x}{y} \right\rangle\Bigg|_{(\pi,4)} \\ &= \left\langle \frac{1}{4}\cos\frac{\pi}{4}, -\frac{\pi}{16}\cos\frac{\pi}{4} \right\rangle \end{aligned}$$

$$= \left\langle \frac{1}{4}\frac{\sqrt{2}}{2}, -\frac{\pi}{16}\frac{\sqrt{2}}{2} \right\rangle$$
$$= \left\langle \frac{\sqrt{2}}{8}, -\frac{\pi\sqrt{2}}{32} \right\rangle$$
$$= \boxed{\frac{\sqrt{2}}{8}\left\langle 1, -\frac{\pi}{4} \right\rangle}$$

d) $f(x,y,z) = x^2 + 2yz - z^2$**,** $(0,1,1)$

$$\nabla f\big|_{(0,1,1)} = \langle f_x, f_y, f_z \rangle\big|_{(0,1,1)}$$
$$= \langle 2x, 2z, 2y - 2z \rangle\big|_{(0,1,1)}$$
$$= \boxed{\langle 0, 2, 0 \rangle}$$

Sinclair, Problem 3

The temperature at a point (x,y) **on a griddle is given by** $f(x,y) = 10 - 2x^2 - y^2$**. Find the maximum rate of change of temperature at** $(2,3)$ ~~**in the direction of** $\langle 1,1 \rangle$~~**.**

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$
$$= \langle -4x, -2y \rangle$$

The maximum rate of change (in any direction) at some point $P(x,y)$ is given by $|\nabla f\big|_{(x,y)}$:

$$|\nabla f\big|_{(2,3)} = |\langle -8, -6 \rangle|$$
$$= \sqrt{(-8)^2 + (-6)^2}$$
$$= \boxed{10}$$

Sinclair, Problem 4

Find the path of an ant which is dropped on a hot griddle at the point $(2,3)$ **if the temperature at a point** (x,y) **is given by** $f(x,y) = 15 - 2x^2 + y^2$**. Assume the ant moves in the coolest direction.**

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$
$$= \langle -4x, 2y \rangle$$

The greatest decrease of f occurs in the direction of $-\nabla f = \langle 4x, -2y \rangle$.

Define a differentiable vector-valued function $\mathbf{r}(t)$ to represent the path the ant travels for $t \geq 0$. The direction of the ant's motion at time t is given by $\mathbf{r}'(t)$. To parametrize the ant's path, set its velocity equal to the negative gradient. This specifies both the direction and a certain speed: $-\nabla f\big|_{\mathbf{r}(t)}$.

Let $x = g(t)$ and $y = h(t)$, so $\mathbf{r}(t) = \langle x, y \rangle = \langle g(t), h(t) \rangle$.

$$\mathbf{r}'(t) = -\nabla f(x,y)$$

$$
\begin{aligned}
&= -\nabla f(\mathbf{r}(t)) \\
&= -\nabla f(x,y)\big|_{\mathbf{r}(t)} \\
&= -\left\langle -4x, 2y \right\rangle\big|_{\mathbf{r}(t)} \\
&= \left\langle 4g(t), -2h(t) \right\rangle
\end{aligned}
$$

Since $\mathbf{r}'(t) = \left\langle g'(t), h'(t) \right\rangle$, we create differential equations in t and solve:

$$
\begin{aligned}
\mathbf{r}'(t) = \left\langle 4g(t), -2h(t) \right\rangle = \left\langle g'(t), h'(t) \right\rangle & \\
&\implies \begin{cases} \dfrac{\mathrm{d}g}{\mathrm{d}t} = 4g(t) \\ \dfrac{\mathrm{d}h}{\mathrm{d}t} = -2h(t) \end{cases} \\
&\implies \begin{cases} g(t) = A_1 e^{4t} \\ h(t) = A_2 e^{-2t} \end{cases}
\end{aligned}
$$

At $t = 0$, $(x, y) = (2, 3)$:

$$
\begin{aligned}
& g(0) = A_1 e^{4(0)} = 2 \\
& h(0) = A_2 e^{-2(0)} = 3 \\
& \quad \implies (A_1, A_2) = (2, 3) \\
& \quad \implies g(t) = 2e^{4t}, h(t) = 3e^{-2t} \\
& \quad \implies \boxed{\mathbf{r}(t) = \left\langle 2e^{4t}, 3e^{-2t} \right\rangle}
\end{aligned}
$$